

07_random_forest

May 3, 2026

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[ ]: import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display

RANDOM_STATE = 42
sns.set_theme(style="whitegrid")
plt.rcParams.update({"figure.dpi": 120})

DATA_DIR = Path("../data")
if not DATA_DIR.exists():
    DATA_DIR = Path("Durga/Lab/MLDL/data")

pd.set_option("display.max_columns", 120)
```

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[2]: from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import ConfusionMatrixDisplay, accuracy_score, \
    classification_report
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import plot_tree

df = pd.read_csv(DATA_DIR / "iris_random_forest.csv")
display(df.head())
display(df["Species"].value_counts().rename_axis("Species").
    reset_index(name="count"))
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

	Species	count
0	Iris-setosa	50
1	Iris-versicolor	50
2	Iris-virginica	50

```
[3]: X = df.drop(columns=["Id", "Species"])
labeler = LabelEncoder()
y = labeler.fit_transform(df["Species"])

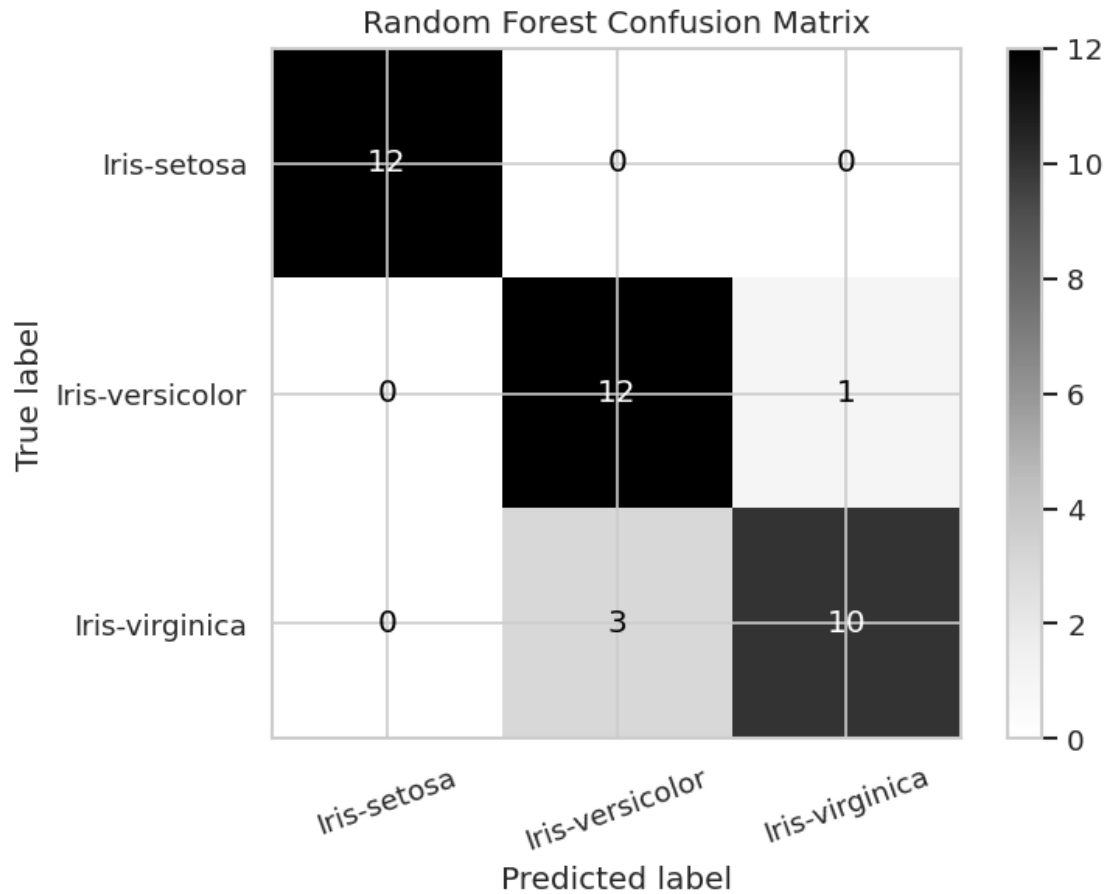
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=RANDOM_STATE, stratify=y
)
forest = RandomForestClassifier(
    n_estimators=200,
    max_depth=5,
    random_state=RANDOM_STATE,
    oob_score=True,
)
forest.fit(X_train, y_train)
pred = forest.predict(X_test)

print(f"Accuracy: {accuracy_score(y_test, pred):.3f}")
print(f"OOB score: {forest.oob_score_: .3f}")
print(classification_report(y_test, pred, target_names=labeler.classes_))
ConfusionMatrixDisplay.from_predictions(y_test, pred, display_labels=labeler.
    ↪classes_, cmap="Greys")
plt.xticks(rotation=20)
plt.title("Random Forest Confusion Matrix")
plt.show()
```

Accuracy: 0.895

OOB score: 0.955

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	12
Iris-versicolor	0.80	0.92	0.86	13
Iris-virginica	0.91	0.77	0.83	13
accuracy			0.89	38
macro avg	0.90	0.90	0.90	38
weighted avg	0.90	0.89	0.89	38



```
[4]: importance = pd.DataFrame(
    {"feature": X.columns, "importance": forest.feature_importances_}
).sort_values("importance", ascending=False)
display(importance)

plt.figure(figsize=(7, 4))
sns.barplot(data=importance, x="importance", y="feature", color="gray",
            edgecolor="black")
plt.title("Random Forest Feature Importance")
plt.show()

plt.figure(figsize=(16, 7))
plot_tree(
    forest.estimators_[0],
    feature_names=X.columns,
    class_names=labeler.classes_,
    filled=True,
    rounded=True,
```

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    max_depth=3,
    fontsize=9,
)
plt.title("One Tree from the Forest")
plt.show()

```

	feature	importance
2	PetalLengthCm	0.456218
3	PetalWidthCm	0.407868
0	SepalLengthCm	0.111614
1	SepalWidthCm	0.024300

